ES 647 Assignment 01

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Problem - Calculate cost of treatment per kg of waste for MSWM facility in India of 500 tonnes of waste per day. The facility follows an integrated approach involving segregation, recycling, composting and refuse derived fuel preparation and disposal of residual inert matter in a sanitary landfill.

Process stage	Typical share of incoming waste	Typical unit cost (₹/tonne)	Daily operating cost for 500 tonne (₹)	Source
Segregation	100 %	1500	7,50,000	CPCB 2021 guidelines data
Recycling	15 %	2000	1,50,000	Swatch Bharat Mission MRF Advisory
Composting	50 %	3000	7,50,000	Typical Aerobic composting - MOHUA, 2022
RDF preparation	20 %	2500	2,50,000	NEERI data for RDF processing
Sanitary Landfilling cost for inert residue	20 %	2000	2,00,000	CPCB landfill O&M - manual 2020
Other Maintenance & overhead costs	—	400	2,00,000	World Bank data 2018 - Municipal overheads.

Total = 23,00,000

⇒ Daily operating cost for the given facility would be around ₹23,00,000. Therefore, cost per kg for the waste treatment would be $\frac{23,00,000}{5,00,000}$ = ₹4.6/kg of waste treated.

~~For comparison, with categories:~~

~~For~~ Comparison with other countries :-

Developed countries like Germany / EU have an implied cost of around 0.08 USD/kg which is around ₹7.1/kg based on EU waste cost study 2022.

USA has an implied cost of about ₹4.8/kg with cost estimations including landfill + operation & maintanance only.

Developing countries like Brazil has an implied cost of waste treatment of ₹3.1/kg based on ~~sources of~~ reports ~~as~~ of Latin American MSWM. Countries like India have average cost of ₹3-5/kg of waste treatment and as shown in estimated for the facility considered here.

Underdeveloped countries like Sub-Saharan Africa spend around ₹1 to 2.50/kg of waste treated based on reports of World Bank without ~~much full cost~~ treatments including facilities like composting, RDF, etc.

Cost Differs Across different categories of countries due to:-

1 - Labour costs - developed economies pay higher wages which implies to higher O&M costs while low income countries befit from lower nominal wages

2 - Technology Adoption - EU/US plants employ costly MRFs, Waste to Energy, ~~cont~~ emission controls while developing regions rely majorly on landfilling / composting.

3 - Environmental Standards - stricter emission and landfill norms in developed nations add compliance cost which most of low or middle income groups does not employ at every facility.

~~4- Logistics - Higher haul costs in dispersed settlements~~

4 - Waste Composition - High moisture / organic content in Indian waste raises treatment cost per tonne while developed countries have proper segregation techniques at the source with high-~~eq~~ tech equipment also available to keep wet ~~moisture~~ waste segregated.

5 - Externalities - Medium or Partial recycling facilities only available due to cost factors ~~are~~ in developing countries with ~~no~~ social cost ~~included~~ unaccounted while in developed countries cost is high due to Automation, technology adoption ~~and~~ with social costs accounted